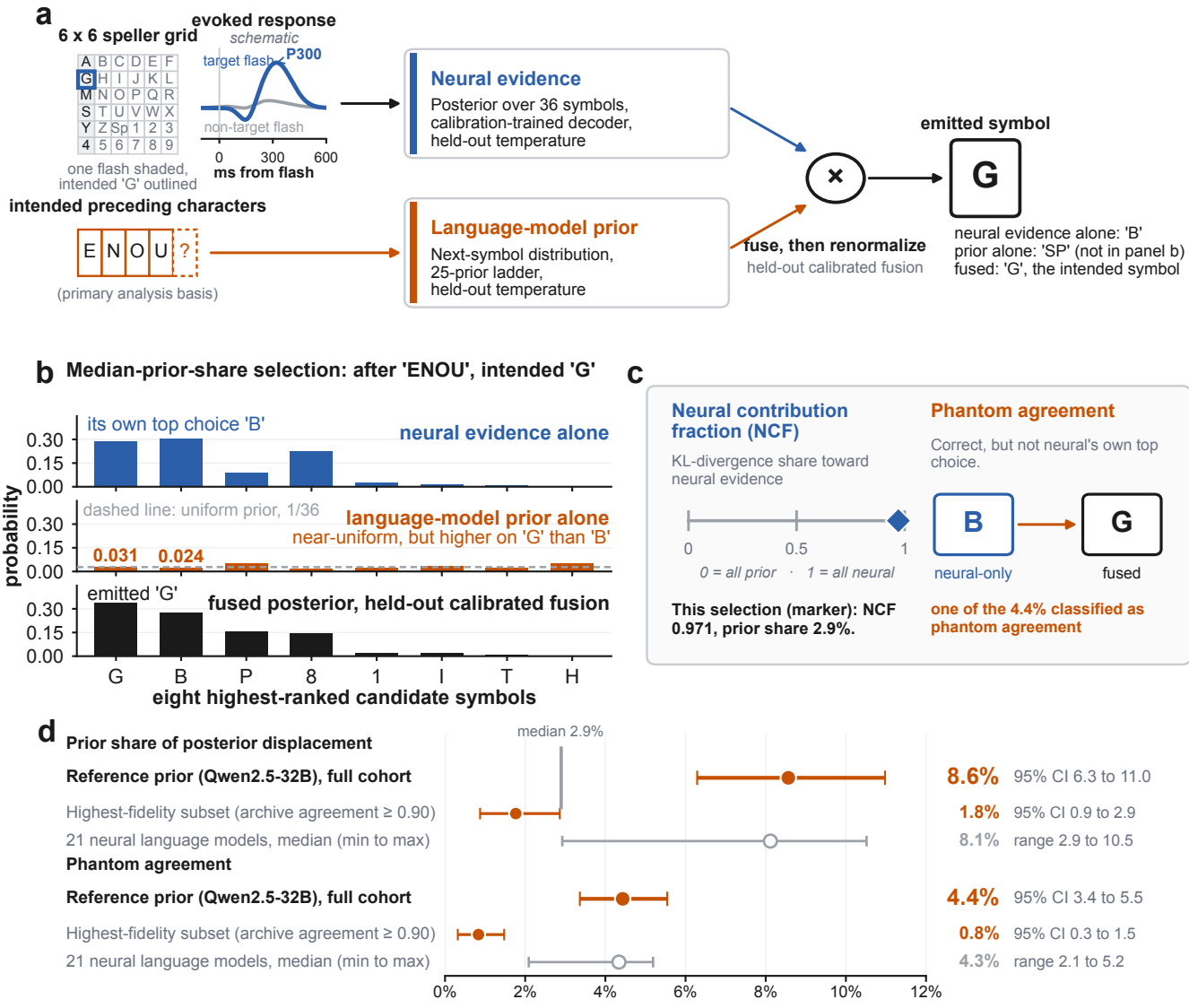

a
6 x 6 speller grid
A B C D E F
G H I J K L
M N O P Q R
S T U V W X
Y Z Sp 1 2 3
4 5 6 7 8 9
one flash shaded, intended 'G' outlined
evoked response
schematic
target flash
P300
non-target flash
0
300
600
ms from flash
Neural evidence
Posterior over 36 symbols, calibration-trained decoder, held-out temperature
intended preceding characters
E N O U ?
(primary analysis basis)
Language-model prior
Next-symbol distribution, 25-prior ladder, held-out temperature
fuse, then renormalize
held-out calibrated fusion
emitted symbol
G
neural evidence alone: 'B'
prior alone: 'SP' (not in panel b)
fused: 'G', the intended symbol
b
Median-prior-share selection: after 'ENOU', intended 'G'
its own top choice 'B'
neural evidence alone
dashed line: uniform prior, 1/36
language-model prior alone
near-uniform, but higher on 'G' than 'B'
0.031
0.024
emitted 'G'
fused posterior, held-out calibrated fusion
probability
0.30
0.15
0.00
G B P 8 1 I T H
eight highest-ranked candidate symbols
c
Neural contribution fraction (NCF)
KL-divergence share toward neural evidence
0
0.5
1
0 = all prior · 1 = all neural
This selection (marker): NCF 0.971, prior share 2.9%.
Phantom agreement
Correct, but not neural's own top choice.
B
G
neural-only
fused
one of the 4.4% classified as phantom agreement
d
Prior share of posterior displacement
median 2.9%
Reference prior (Qwen2.5-32B), full cohort
8.6% 95% CI 6.3 to 11.0
Highest-fidelity subset (archive agreement ≥ 0.90)
1.8% 95% CI 0.9 to 2.9
21 neural language models, median (min to max)
8.1% range 2.9 to 10.5
Phantom agreement
Reference prior (Qwen2.5-32B), full cohort
4.4% 95% CI 3.4 to 5.5
Highest-fidelity subset (archive agreement ≥ 0.90)
0.8% 95% CI 0.3 to 1.5
21 neural language models, median (min to max)
4.3% range 2.1 to 5.2
0%
2%
4%
6%
8%
10%
12%